



MODERN EDUCATION SOCIETY'S

NOWROSJEE WADIA COLLEGE

(An Autonomous College under Savitribai Phule Pune University)

Course Code:PCSMJ123

Laboratory Course on Web Frameworks

National Education Policy (NEP) Syllabus

To be implemented from Academic Year 2023-2024

M.Sc.(Computer Science)-Part I

Semester II

2025-26

Co-Ordinator

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Prepared by

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Roll No :

Name :

**WEB FRAMEWORKS
ASSIGNMENT COMPLETION SHEET**

Sr. No	Assignment Name	Marks
1	Javascript	
2	Node JS	
3	MongoDB	
4	Express JS Framework	
5	DJango Framework	
Total marks out of 25		
Total marks out of 10		

Signature of Batch In-charge:

Internal Examiner :

External Examiner :

Date:

Assignment 1: JavaScript

Pre-Reading

- Javascript Basics
- Regular Expressions
- Functions, Array

Lab Assignments

1. Create an HTML form that contains the Student Registration details and write a JavaScript to validate Student first and last name as it should not contain other than alphabets and age should be between 18 to 50. If the form inputs are valid, display a success message; otherwise, show error messages next to the relevant fields.
2. Create an HTML form that contains the Employee Registration details (Employee Number, Employee Name, Salary, DOB, Joining Date.) and Write a JavaScript to validate DOB, Joining Date i.e. DOB should be less than the Joining Date by 20 Years.
3. Design a JavaScript application that includes an HTML login form where users enter their Username, Email, and Password. Implement JavaScript validation using Regular Expressions to ensure the Username and Email meet the required format. Display an error message if the username, email format or password is invalid.

Extra Practice Question:

Create an HTML form that contains the Admission details such as PRN No. Name, Class, previous qualification details. and Write a JavaScript to validate PRN No. such that it should not exceed 12 digits and only numbers must be allowed for that field. Also convert Name to uppercase on-keyup event.

Signature of the instructor Date

Assignment Evaluation

0: Not done	<input data-bbox="464 1759 540 1833" type="checkbox"/>	2: Late Complete	<input data-bbox="933 1759 1010 1833" type="checkbox"/>	4: Complete	<input data-bbox="1338 1759 1414 1833" type="checkbox"/>
1: Incomplete	<input data-bbox="464 1854 540 1927" type="checkbox"/>	3: Needs Improvement	<input data-bbox="933 1854 1010 1927" type="checkbox"/>	5: Well Done	<input data-bbox="1338 1854 1414 1927" type="checkbox"/>

Assignment 2: Node JS

Pre-Reading

- File System Module and HTTP Module
- Events

Lab Assignments

1. Using nodejs create a web page to read two file names from the user and append contents of the first file into the second file and Delete the first file.
2. Write node js script to build Your own Node.js Module. Use require ('http') module is a built-in Node module that invokes the functionality of the HTTP library to create a local server. Also use the export statement to make functions in your module available externally. Create a new text file to contain the functions in your module called, "modules.js" and add this function to return today's date and time.
3. Create a js file named main.js for an event-driven application. There should be a main loop that listens for events, and then triggers a callback function when one of those events is detected.

Extra Practice Question:

1. Design a Node.js application that implements a module named circle.js to calculate the area and circumference of a circle. The application should export the functions area() and circumference() from the module and import them into a separate script named app.js. Implement logic to use these functions for a given radius and display the results in the console, ensuring the application includes validation to handle invalid inputs such as negative values or non-numeric data.
2. Design a Node.js application that reads a file and counts the occurrences of a user-specified word in a case-insensitive manner using the File System module. The application must display the word count in the console and include error handling to manage missing files or invalid inputs.

Signature of the instructor

Date

Assignment Evaluation

0: Not done

☐

2: Late Complete

☐

4: Complete

☐

1: Incomplete

☐

3: Needs Improvement

☐

5: Well Done

☐

Assignment 3: MongoDB

Pre-Reading

- MongoDB CRUD Operations
- Methods for CRUD

Lab Assignments

1. Design a Node.js application that establishes a connection to a MongoDB database, creates a collection named “customers”, and inserts five customer records into it. The application should retrieve all records from the collection, store them in a result object, and display the results on console.
2. Design a Node.js application that establishes a connection to a database to insert multiple records into a “student” table and displays the resulting object on the console. The application must implement robust error handling to manage connection failures or query issues.
3. Create a node.js file that Select all records from the “customers” table, and delete a specific record based on a given condition.

Signature of the instructor

Date

Assignment Evaluation

0: Not done

2: Late Complete

4: Complete

1: Incomplete

3: Needs Improvement

5: Well Done

Assignment 4: Express JS

Pre-Reading

- Routing and Views
- Rendering Data

Lab Assignments

1. Create User Login System using Express JS containing login form details i.e. username and password. and also Validate username such that it should not contain more than 6 characters in it.
2. Develop a Node.js application using Express.js and EJS to build a dynamic recipe book featuring multiple pages like Home and Recipe Details. The application must implement routing to render EJS templates, pass dynamic data to views, and provide functionality to add and delete recipes through dedicated POST and DELETE routes. Ensure the system manages recipe storage effectively and handles user interactions.
3. Develop a Node.js web application using Express.js that allows users to download a file as an attachment. The application should configure an Express.js server to handle file transfer requests and provide an endpoint that prompts the browser to download the file. Include an HTML page with a Download button to trigger the transfer, and implement proper error handling to manage missing or incorrect files.

Extra Practice Question:

Using Node.JS, create a web page (Use EJS Template Engine) to accept the employee details and render that page on the browser. Process the filled data using the POST method and display the data on the next page.

Signature of the instructor

Date

Assignment Evaluation

0: Not done

2: Late Complete

4: Complete

1: Incomplete

3: Needs Improvement

5: Well Done

Assignment 5: Django Framework

Pre-Reading

- About 3 Core files(model.py,urls.py,views.py) and URL Patterns
- Views And Templates
- Forms & Validations

Lab Assignments

1. Design Django application to display “Hello,Welcome to School of Computer science at NWC” message using template.
2. Design a Django application that adds web pages with views and templates.
3. Implement URL mapping and page redirection by using django frameworks which contain at least two pages.
4. Create a login application using the django framework on successful login display appropriate message.Ensure proper error handling for incorrect credentials and display an error message.

Extra Practice Questions:

1. Design django form which contains student details (fname,lname,class,email, MobNo,address,resume_file) on submit button display student details on web page.Implement form validation to ensure that the email and mobile number are correctly formatted and the class field only accepts predefined values.The form should also restrict the uploaded resume to PDF or DOCX formats with a maximum file size limit.
2. Create a django application which contains HomePage and AboutUs pages.whenever it is redirected to about_us function it displays “An Autonomous College under Savitribai Phule Pune University” message otherwise it return response as “Welcome to Nowrosjee Wadia College ,Pune”.

Signature of the instructor

Date

Assignment Evaluation

0: Not done

2: Late Complete

4: Complete

1: Incomplete

3: Needs Improvement

5: Well Done